Product Requirement Document (PRD)

📱 Project Name: Healthify
Version: 1.0.0
Status: Draft
Target Launch: Q2 2027
Platform: Cross-Platform Mobile (iOS & Android) via React Native

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1. Executive Summary & Goals
1.1 Purpose
Healthify is a privacy-first, AI-driven fitness and healthcare companion application. It bridges the gap between raw health metrics (tracked via wearables) and actionable, highly engaging user habits by utilizing conversational AI insights, interactive 3D exercise visualizations, and deeply personalized meal metrics.

1.2 Core Objectives
- **Engage:** Provide highly visual, 3D-animated instruction to lower the barrier of entry for home/gym workouts.
- **Personalize:** Tailor holistic nutrition guides accommodating cultural, ethical, and medical dietary requirements.
- **Simplify:** Translate complex health data into scannable, interactive summaries powered by Google Gemini API.

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2. Target Audience & Personas
- **The Visual Learner (Fitness Novice):** Needs precise, interactive guidance on workout forms to prevent injury without hiring a personal trainer.
- **The Strict Dieter (Specific Macro Requirements):** Requires granular control over Veg/Non-Veg/Medical meal categorization with automated daily planning.
- **The Data-Driven Wellness Enthusiast:** Wants high-level health correlations (e.g., "How does my sleep affect my workout performance?") without reading tabular logs.

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3. Product Architecture & Technical Stack
- **Frontend Framework:** React Native (Expo Workflow) with TypeScript.
- **Styling:** NativeWind (TailwindCSS) configured with a strict Dark Mode palette.
- **AI Engine:** Google Gemini API (via Google AI Studio integration).
- **3D Graphics:** Expo-GL / Three.js (React Three Fiber) or optimized glTF/Lottie webview loaders for low-overhead mobile 3D animations.
- **State Management:** Zustand or Redux Toolkit.

- **Database/Backend:** Firebase (Authentication & Firestore) or Supabase.

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4. Functional Requirements

🛠️ Core Module 1: System-Wide Dark Mode UI
* **FR-1.1:** The app must natively support and enforce a Dark Mode system, overriding system light preferences if necessary.
* **FR-1.2:** Design Tokens:
  * Surface Background: `#121212`
  * Card Surfaces: `#1E1E1E`
  * Accents: Neon Green (`#00E676`) for Fitness/Go actions; Neon Blue (`#29B6F6`) for Data/AI insights.

🤖 Core Module 2: Interactive AI Health & Exercise Summary
* **FR-2.1:** The app must parse aggregated user telemetry (Steps, Calories, Active Minutes, Sleep hours) and feed it safely to the Gemini API.
* **FR-2.2:** The dashboard must display an interactive text box/card showing a **2-sentence summary** of the user's current health status.
* **FR-2.3:** The component must include an interactive "Ask Healthify AI" chat drawer for dynamic health and exercise queries.

🏋️ Core Module 3: 3D Exercise Demonstration Center
* **FR-3.1:** The workout module must categorize exercises into explicit tabs: **Upper Body**, **Core**, **Lower Body**, and **Cardio**.
* **FR-3.2:** Selecting an exercise must launch a focused video/3D engine viewport that loops a short 3D animation detailing correct posture and muscular engagement.
* **FR-3.3:** The 3D viewport must support touch gestures (rotation and zoom) so users can examine the form from multiple angles.

🥗 Core Module 4: Tailored Meal Planner & Diet Guide
* **FR-4.1:** Upon onboarding, users must select explicit dietary profiles: **Veg**, **Non-Veg**, **Vegan**, **Keto**, or **Custom Medical**.
* **FR-4.2:** The app must generate a dynamic 7-day caloric plan matching the selected tag.
* **FR-4.3:** The UI must display an interactive Macro-Nutrient ring breakdown (Protein, Carbs, Fats) with tap-to-log functionality.

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5. Non-Functional Requirements

🔒 Security & Privacy
- **NFR-5.1:** All health data must be encrypted at rest and in transit.

- **NFR-5.2:** Personal Health Information (PHI) passed to the Gemini API must be anonymized (stripped of names, emails, and exact locations).

⚡ Performance & Scalability
- **NFR-5.3:** 3D animation assets must be compressed to `.glb` format under 2MB per file to prevent memory leaks and frames-per-second (FPS) drops below 60fps on mid-range devices.
- **NFR-5.4:** AI latency responses for health summaries must not exceed 2.5 seconds.

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6. Release & Phase Plan

